Boxuan Wang / Imperial Design Engineering

S00

Humanoid torso structure

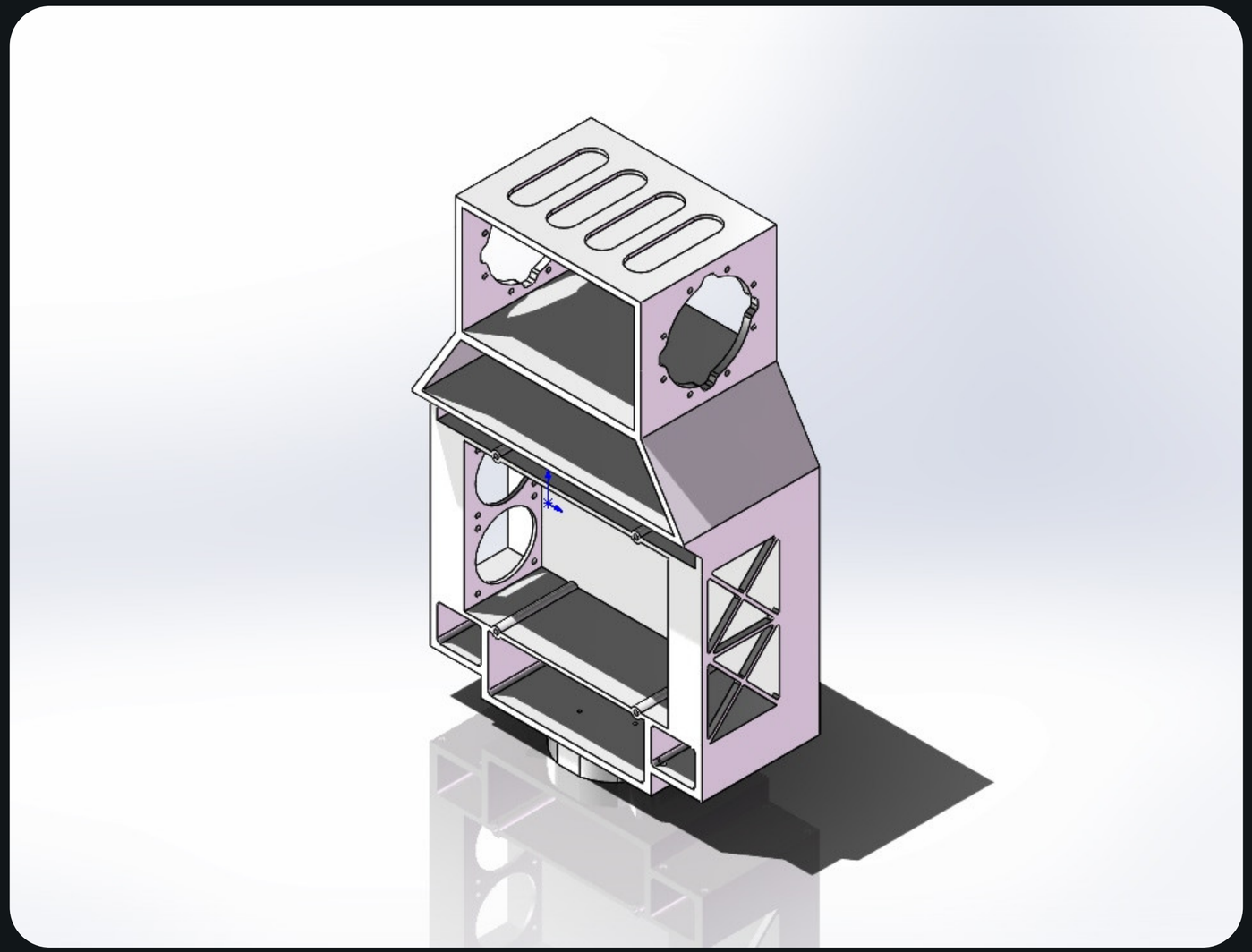
Humanoid Development Intern
July-September 2026

I designed connector parts and structural supports in SolidWorks, and contributed to the team's redesign of upper-torso load paths.

18%

Lighter torso
Team redesign · simulation

Team redesign result. Fatigue requirements met in Ansys simulations with a safety factor of 3.



Saved CAD revision. The team simulation result does not validate this pictured configuration.

jw4424@ic.ac.uk mystify59.github.io

Package, seal and assemble.

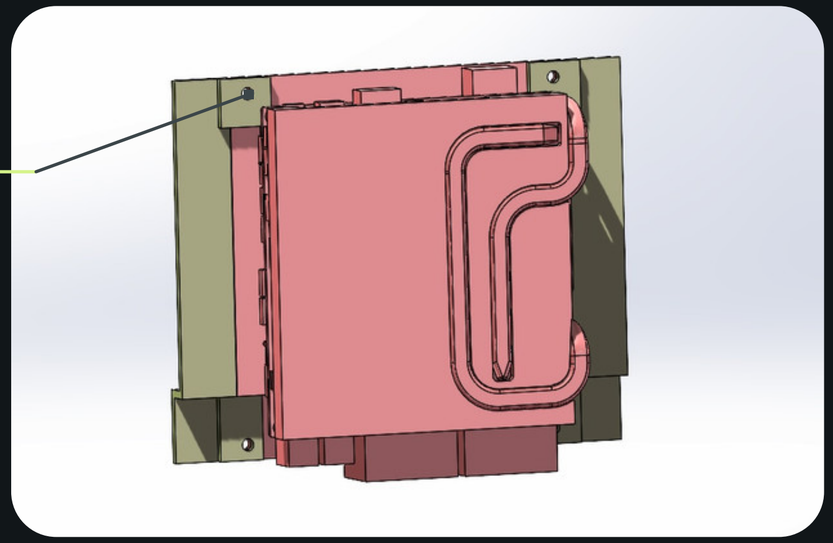
Fit the CCU and air duct into the torso while keeping neighbouring components clear.

Two identical parts

I divided the sealing structure into paired parts with matching M4 interfaces in the main frame.

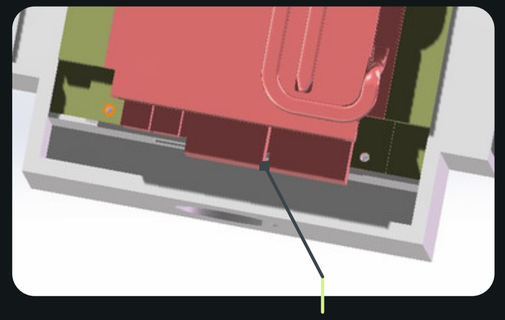
Guide the assembly

I allowed room for adhesive-foam seals and added lower locating grooves to guide installation.





Main-frame mounting envelope



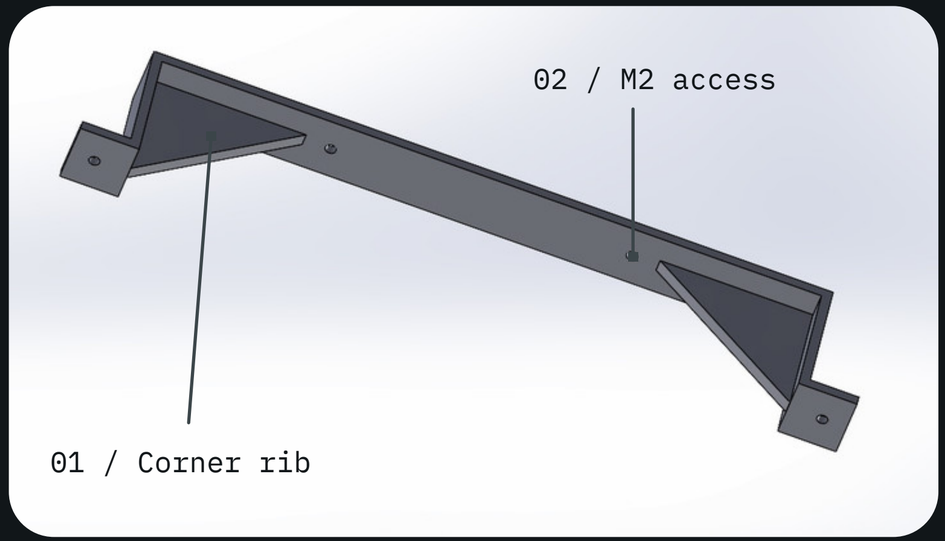
Original CAD: component packaging, interfaces and assembly clearances.

Refine the microphone support.

Separate the microphone support from the camera bracket and mount it directly to the main structure.

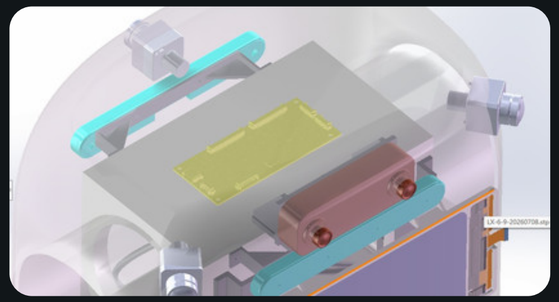
Reinforce the longer span

The longer span increased the lever arm. I added corner ribs while keeping M2 fixing access clear.

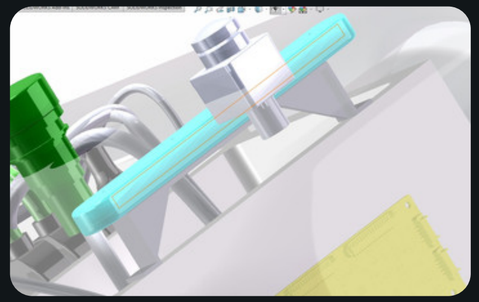


Check the surrounding assembly

I reviewed component geometry, mounting access and the neighbouring assembly in CAD.



Assembly context



Mounted detail

CAD development evidence. Physical performance of this revision remains unverified.

Aquatorium

A compact system for live-fish transport

ACADEMIC TEAM PROJECT

The team developed a collapsible carrier with individual fish bags, a support frame, oxygen delivery and cooling.

My Phase 1 / concept exploration

I explored six early concepts, including a floating hydro-generator. Later development shown here is team work.

22 L → 8 L

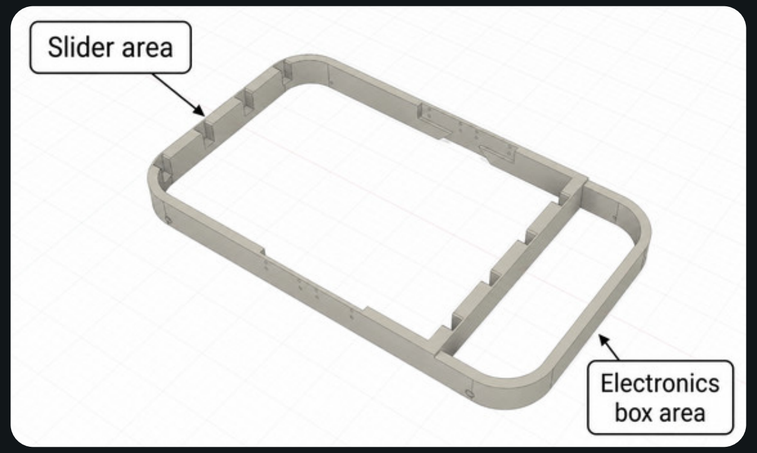
Reported team result: unfolded / folded volume.



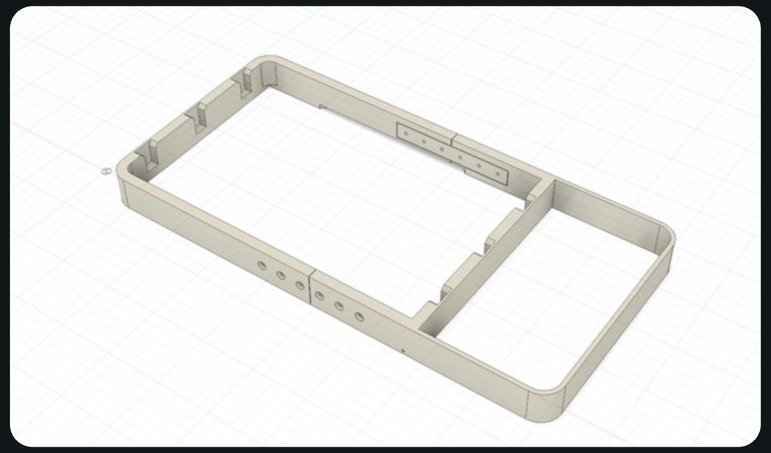
Original concept visualisation from the academic team project.

Iterate around the components.

The team revised the frame and built three enclosure iterations around the electronics and access requirements.



01 / Earlier frame



02 / Revised frame

4 → 3 bag bays

Fewer bays made space for the electronics and improved slider access.

Three iterations

Basic enclosure → test layout → refined openings and mounts.



01 / Basic enclosure

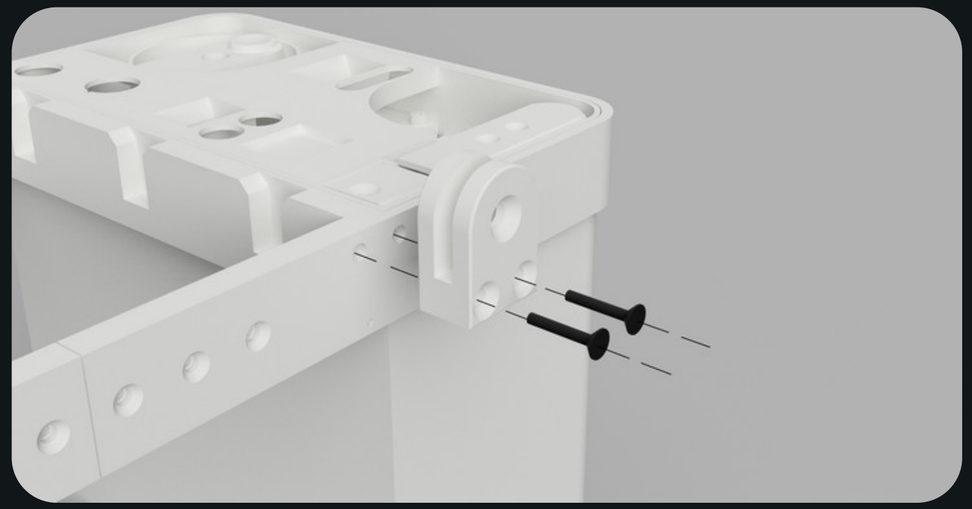


02 / Test layout

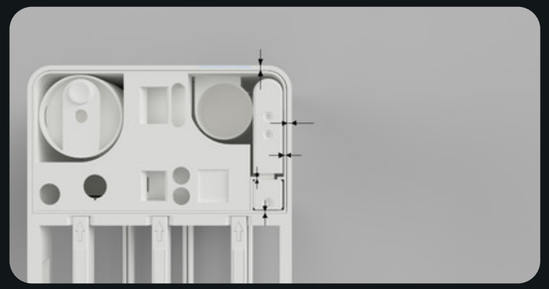


03 / Refined openings

Test the prototype. Plan the product.



Proposed subassemblies and service access.



Manufacturing study



Collapsed concept render

About 4 hours

Reported prototype cooling runtime. Battery endurance, rain protection, charging and carrying weight remained priorities.

Design for service

The team proposed 2 mm shell and 3 mm support walls, removable bag/electronics modules and standard screw fixings.

Next iteration

Longer runtime and manufacturing readiness remain design goals rather than verified outcomes.

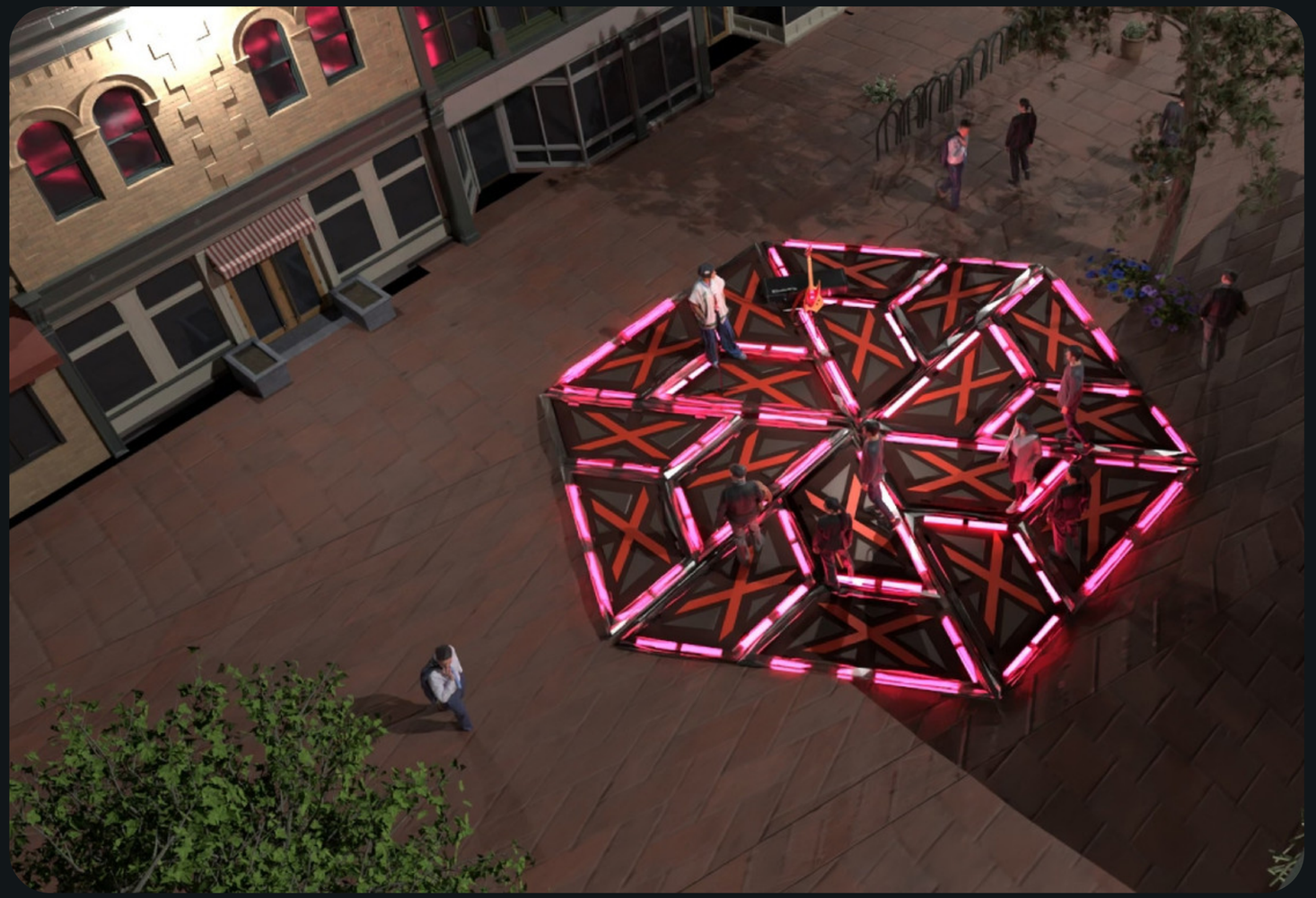
Street Pulse

Feel the performance

Modular floor units combine vibration and lighting to invite audiences into street performances.

Research, make and test

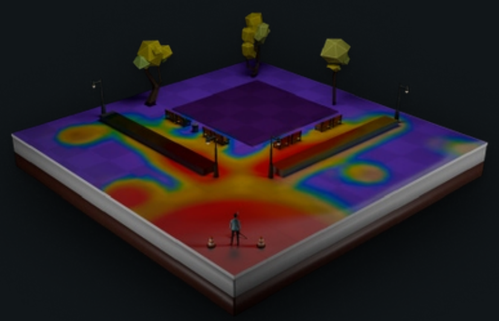
Almost all user research; tessellation and exterior design; electronics prototyping, testing and assembly.



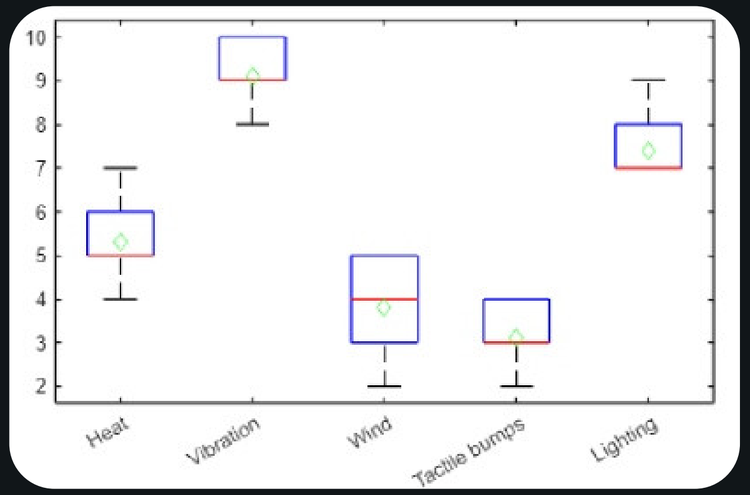
Academic team project, JJAGernaut. Larger platform concept render; the team built one physical demonstration module.

Understand both sides of the performance.

I carried out almost all user research, using observation, interviews, personas and co-design.



Perceived pleasantness / 1-10 scale



n = 10 recorded ratings; verbal endpoints not stated.

Audience + performer

The team compared brief audience encounters with performers' full working days. Density mapping described how people gathered.

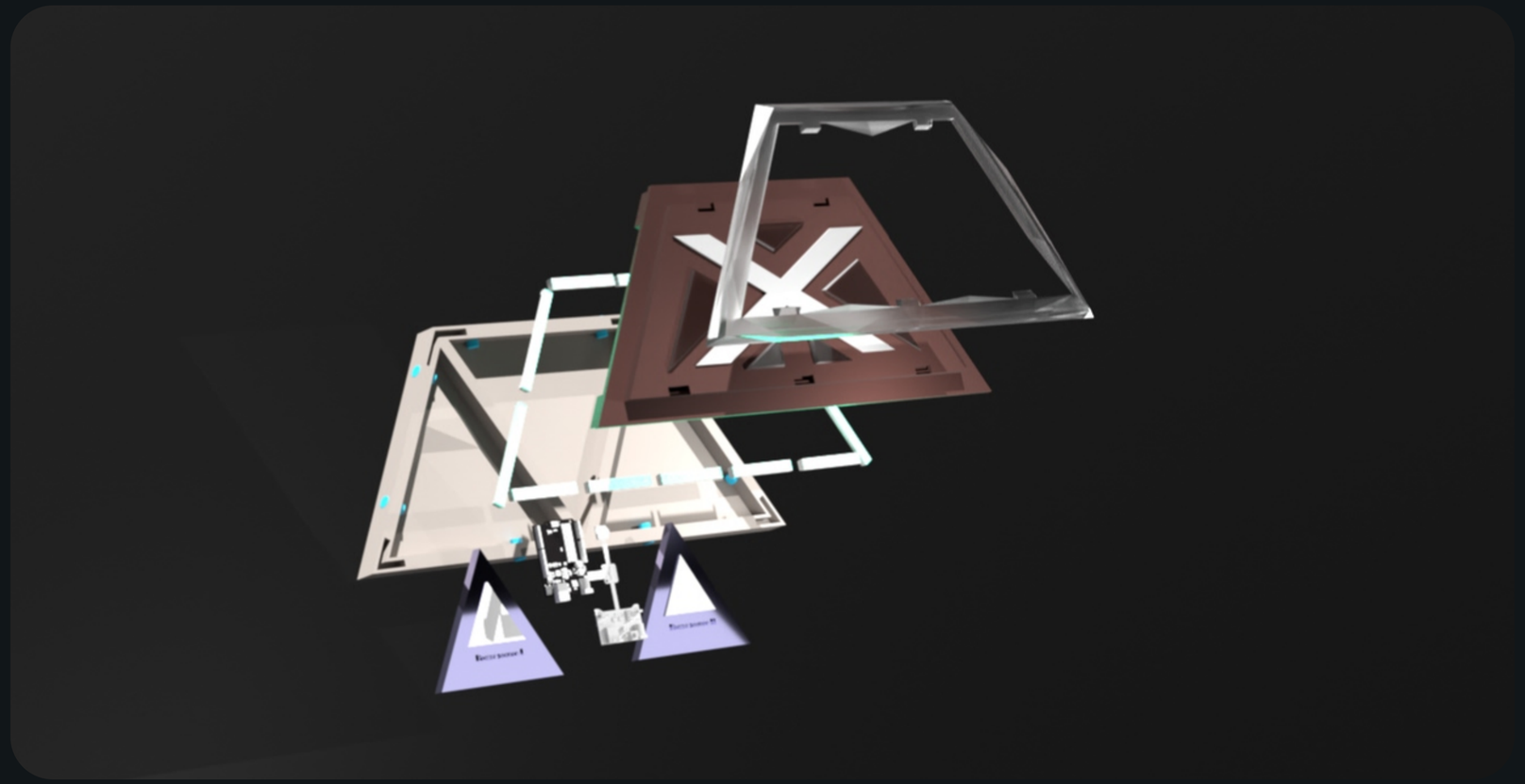
Ground haptics + light

Vibration had the highest mean (9.1/10), followed by optics (7.4/10). The team pursued vibration and light.

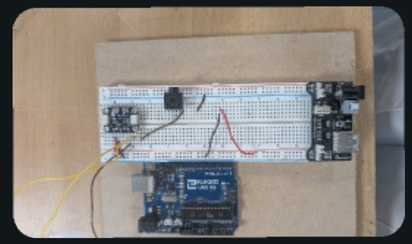
Original team research. Left: qualitative density illustration. Right: report Figure 1.5; ten recorded entries.

Prototype. Test. Refine.

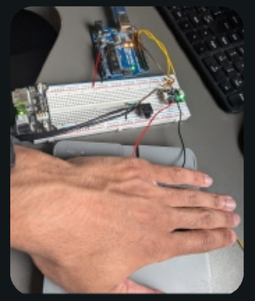
I prototyped the electronics and carried out all testing and assembly. Other team members developed the software.



Exploded module concept. The team built one physical demonstration module.



Breadboard prototype



Surface test

3 mm acrylic

Selected for vibration transmission.

Firmer feet

Replaced soft mats after actuators touched the floor.

19-person virtual evaluation

52.6% agreed and 36.8% strongly agreed that it improved the viewing experience.

One working module with a virtual performance; no live-street trial.

Project Revival

From screen to physical form

A personal series of film- and game-inspired builds, explored through CAD, printable parts and assembly.

Anvil / ARC Raiders

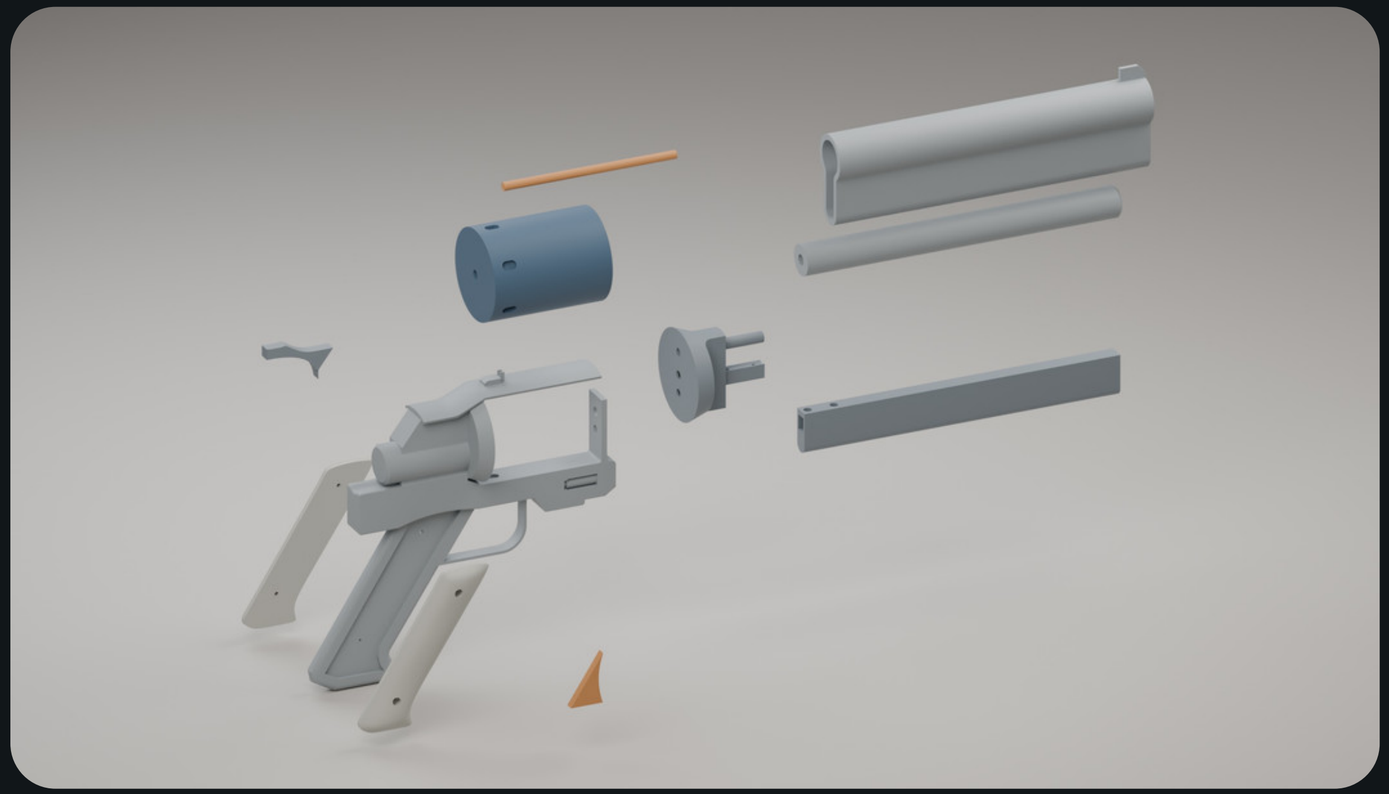
I created the initial Fusion model and component architecture. The first full-scale iteration was printed in China.



Original CAD with a proposed metal finish. This is a finish visualisation, not a photograph of the print.

From one form to eleven components.

Break a complex screen reference into printable components and an assembly to develop and test.



Original eleven-component CAD assembly. Appearance reference: ARC Raiders.

Separate the assembly

Body, outer details, grip panels, chamber and central shaft are organised as distinct components.

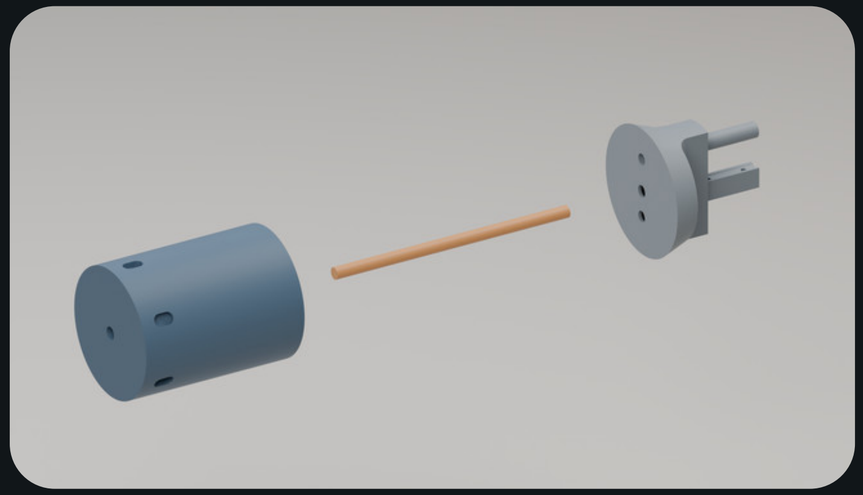
Check fit through making

The first iteration was printed. Internals remain in development; fit, movement and sound require physical testing.

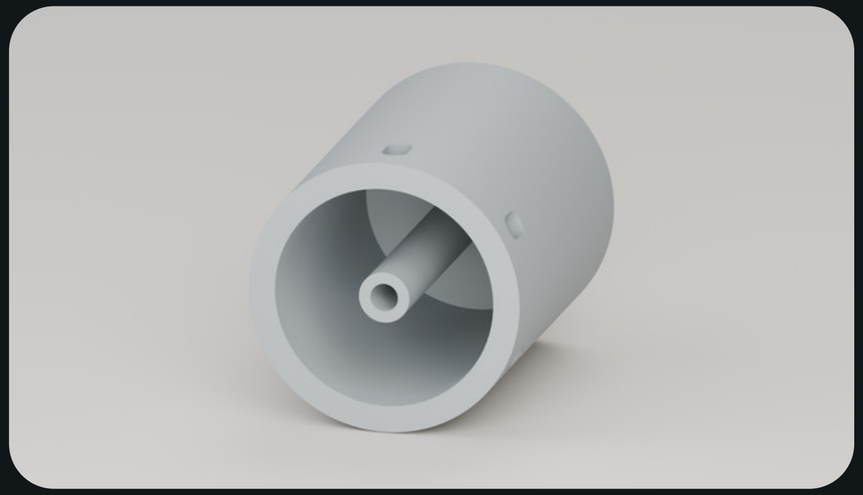
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Develop the next iteration.

Anvil internals remain in development, pending completion and physical tests of fit, movement and sound.



Original Fusion chamber-and-shaft study.



Anvil decorative-drum study · CAD only · Codex/Fusion assisted.

ANVIL / INTERNAL DEVELOPMENT

The first iteration was printed. The internals are still being developed; a working prop or mechanism is not yet established.

Ghost / planned companion

The Destiny-inspired companion has a staged hardware and voice-system plan. It remains in development.